

Scale-Free Common-Container Comparisons Do Not Entail Numerical Inequalities

A Minimum-Viable No-Go Theorem Motivated by the IUT III Theorem 3.11-to-Corollary 3.12 Bridge

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Abstract

Motivated by the formalization problem surrounding the passage from Theorem 3.11 to Corollary 3.12 in Mochizuki’s *Inter-universal Teichmüller Theory III*, we study a deliberately modest typed abstraction of a common-container-to-numerical bridge. The abstraction contains two typed log-volume domains, nontrivial monoid and hull objects, log-volume extraction maps, a common container, embeddings into that container, a container comparison certificate, and final numerical evaluation maps. It deliberately excludes any bridge certificate such as a common numerical chart, a scale-preserving transport map, a certified relation, interval-valued quotient data, sandwich estimates, bounded-distortion margin data, normalized monoid/hull valuations, proof-path ranks, or calibration plus comparison-realization.

We prove a minimum-viable no-go theorem: this scale-free monoid/hull/log-volume/common-container abstraction does not entail the corresponding numerical ordered-real inequality. The proof is by an active relative rescaling countermodel. In the model, the common-container comparison holds, while the final numerical inequality fails. We then classify the minimal “escape hatches” that would defeat the countermodel. These include chart bridges, transport bridges, certified relation bridges, interval bridges, sandwich bridges, bounded-distortion-plus-margin bridges, normalized valuation bridges, proof-path-rank bridges, and calibration-plus-comparison-realizer bridges.

The result is intentionally narrow. It is not a proof or disproof of IUT. It is not a claim that the actual IUT proof lacks the required bridge data. Its intended contribution is methodological: it gives a formal diagnostic lower bound on what any successful common-container-to-numerical bridge must contain. In particular, if a proposed formalization of a Theorem 3.11-to-Corollary 3.12-like passage contains only scale-free monoid/hull/log-volume/common-container comparison data, then that abstraction is too weak to imply the numerical inequality.

Keywords: Inter-universal Teichmüller theory; Corollary 3.12; common containers; log-volume; formalization; typed abstraction; bridge certificates; no-go theorem; scale retention; proof auditing.

Mathematics Subject Classification (2020): 11G99, 03B30, 03B35, 11J25

Subject Extraction: A formal diagnostic study of the logical bridge between common-container comparisons and numerical ordered-real inequalities, specifically addressing the transition from Theorem 3.11 to Corollary 3.12 in IUT III.

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1 Introduction

The purpose of this note is to isolate a small formal pattern that occurs in discussions of the final Diophantine step of IUT III: a comparison is obtained inside a common container, and a numerical ordered-real inequality is then desired. The local IUT-motivating passage is the implication from Theorem 3.11 to Corollary 3.12 in IUT III. This passage is significant enough that Mochizuki’s April 2026 note on formalization identifies it as Stage 1 of a proposed LeanForm strategy [1].

This note does not attempt to formalize IUT. It does not attempt to interpret all of Theorem 3.11 or Corollary 3.12. It does not prove that Mochizuki’s argument succeeds, nor does it prove that the argument fails. The aim is much smaller: to prove a no-go theorem for a deliberately specified abstraction that is too weak to contain a scale-retaining bridge.

The core abstract pattern is this. There are two typed log-volume domains,

$$S_q, \quad S_\Theta,$$

with distinguished elements

$$\ell_q \in S_q, \quad \ell_\Theta \in S_\Theta.$$

There is a common container C , with maps

$$\iota_q : S_q \rightarrow C, \quad \iota_\Theta : S_\Theta \rightarrow C,$$

and a comparison certificate

$$\pi : \iota_q(\ell_q) \preceq_C \iota_\Theta(\ell_\Theta).$$

There are also numerical evaluation maps

$$F_q : S_q \rightarrow \mathbb{R}, \quad F_\Theta : S_\Theta \rightarrow \mathbb{R}.$$

The desired numerical conclusion is

$$\text{Num} := F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

The central question is whether the container comparison π entails Num . This note proves that the answer is no for a scale-free abstraction A_{MHL} containing monoid, hull, log-volume, common-container, and numerical-evaluation structure but no bridge certificate. The no-go theorem is witnessed by a concrete countermodel in which

$$\iota_q(\ell_q) \preceq_C \iota_\Theta(\ell_\Theta)$$

holds, but

$$F_q(\ell_q) \leq F_\Theta(\ell_\Theta)$$

fails.

The relevance to IUT is deliberately modest. IUT III itself discusses multiradial algorithms, splitting monoids of LGP-monoids, and log-volume estimates [3]. Scholze and Stix’s widely discussed report criticizes the passage to a numerical inequality by focusing, among other things, on the need to spell out identifications among copies of ordered one-dimensional real-vector spaces when comparing real numbers [2]. This paper extracts a formal lower-bound diagnostic from that issue: if a formal abstraction does not contain a bridge certificate that connects common-container comparison to numerical evaluation, then the numerical inequality is not entailed.

2 What this note does and does not claim

2.1 Claims

This note claims the following.

First, we define a typed abstraction A_{MHL} containing:

monoid data+hull data+log-volume maps+common-container maps+container comparison+numerical maps.

Second, we prove:

$$A_{\text{MHL}} + \pi \not\models F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

Third, we classify extra bridge mechanisms that would defeat the countermodel.

2.2 Non-claims

This note does not claim:

- (i) that IUT is false;
- (ii) that IUT III Corollary 3.12 is false;
- (iii) that Mochizuki’s proof lacks a bridge certificate;
- (iv) that Scholze–Stix’s critique is formally complete;
- (v) that the abstraction A_{MHL} is a faithful model of IUT;
- (vi) that the countermodel is a model of any actual Hodge theater, Frobenioid, theta-link, or multiradial algorithm.

The note claims only that one particular scale-free abstraction is insufficient.

2.3 Why this modest result may still be useful

The result may be useful because it clarifies a formalization target. If a formal proof attempts to move from a common-container/log-volume comparison to a numerical ordered-real inequality, then it must supply some additional bridge data. Possible forms include:

$$\kappa, \quad g, \quad R_B, \quad K, \quad \Phi, \quad A/\delta/\varepsilon, \quad V, \quad \text{RankPath}.$$

Informally, these stand for a numerical chart, a scale-preserving transport, a certified relation, an interval-valued quotient bridge, sandwich estimates, bounded-distortion margin data, a normalized valuation, and a proof-path rank.

3 Background and motivation

This section gives only the minimal public context needed to motivate the abstraction.

Mochizuki’s IUT III contains the theorem/corollary region that motivates the present abstraction. IUT III discusses multiradial algorithms, splitting monoids of LGP-monoids, and log-volume estimates, and identifies Corollary 3.12 as part of the relevant log-volume/Diophantine transition [3]. Scholze and Stix’s report criticizes the passage to a numerical inequality by focusing, among other things, on the need to spell out identifications among copies of ordered one-dimensional

real-vector spaces when comparing real numbers [2]. Mochizuki’s April 2026 formalization note identifies the implication [IUTchIII] Theorem 3.11 $\Rightarrow$ Corollary 3.12 as the first stage of a proposed LeanForm strategy [1].

Dupuy and Hilado’s work on the statement of Corollary 3.12 is another example of focusing attention on the local Corollary 3.12 region, although their paper studies computability of initial theta data rather than the specific no-go abstraction studied here [4].

This note should be read as a small formal-methods contribution inspired by that local setting. Its concern is not the arithmetic content of IUT, but the typed proof pattern

$$\text{common-container comparison} \Longrightarrow \text{numerical ordered-real inequality}.$$

4 The bridge-certificate viewpoint

The central bridge pattern may be written as

$$\pi \Rightarrow R_B(\ell_q, \ell_\Theta, \pi) \Rightarrow F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

Here:

- π is the comparison certificate in the common container;
- R_B is a bridge relation supplied by a certificate B ;
- the first implication is *extraction*;
- the second implication is *numerical soundness*.

The conditional theorem is trivial but important.

Proposition 4.1 (Conditional bridge theorem). *Suppose there exists a relation $R_B(x, y, \rho)$ such that*

$$\pi \Rightarrow R_B(\ell_q, \ell_\Theta, \pi)$$

and

$$R_B(x, y, \rho) \Rightarrow F_q(x) \leq F_\Theta(y).$$

Then

$$F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

Proof. Apply extraction to π , obtaining

$$R_B(\ell_q, \ell_\Theta, \pi).$$

Then apply numerical soundness with $x = \ell_q$, $y = \ell_\Theta$, and $\rho = \pi$. □

The conditional theorem is not the substantive work. The substantive work is constructing B non-circularly. A bridge certificate B should satisfy four conditions:

- (i) *extractability*: B is obtained from the comparison proof π ;
- (ii) *numerical soundness*: B implies the desired ordered-real inequality;
- (iii) *non-circularity*: B is not merely the desired conclusion renamed;
- (iv) *scale-robustness*: B blocks or neutralizes active relative rescaling.

The no-go theorem below proves that A_{MHL} lacks such a bridge.

5 Formal abstraction A_{MHL}

The subscript “MHL” stands for *monoid/hull/log-volume*.

5.1 Typed domains

There are two typed log-volume domains

$$S_q, \quad S_\Theta.$$

Even if both are represented by copies of $\mathbb{R}$, they are treated as distinct types. Hence an expression such as

$$\ell_q \leq \ell_\Theta$$

is not part of the formal language.

5.2 Pilot, monoid, hull, and log-volume data

For each $i \in \{q, \Theta\}$, there are:

- a pilot domain P_i ;
- a commutative monoid M_i ;
- a hull domain H_i ;
- a map $\text{Mon}_i : P_i \rightarrow M_i$;
- a map $\text{Hull}_i : M_i \rightarrow H_i$;
- a map $\text{LogVol}_i : H_i \rightarrow S_i$;
- a distinguished pilot $p_i \in P_i$.

The distinguished log-volume element is

$$\ell_i = \text{LogVol}_i(\text{Hull}_i(\text{Mon}_i(p_i))).$$

5.3 Common-container data

There is a common container C , maps

$$\iota_q : S_q \rightarrow C, \quad \iota_\Theta : S_\Theta \rightarrow C,$$

and a relation

$$\preceq_C \subseteq C \times C.$$

There is a comparison certificate

$$\pi : \iota_q(\ell_q) \preceq_C \iota_\Theta(\ell_\Theta).$$

5.4 Numerical data

There are numerical evaluation maps

$$F_q : S_q \rightarrow \mathbb{R}, \quad F_\Theta : S_\Theta \rightarrow \mathbb{R}.$$

The target numerical conclusion is

$$\text{Num} := F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

5.5 Deliberately excluded bridge data

The abstraction A_{MHL} does not include any of the following:

$$R_B, \quad \kappa, \quad g, \quad K, \quad \Phi_q, \Phi_\Theta, \quad A_q, A_\Theta, \delta, \varepsilon_q, \varepsilon_\Theta,$$

nor a normalized monoid/hull valuation, a proof-path rank, or a calibration-plus-comparison-realizer.

This exclusion is not intended to mimic actual IUT. It is intended to define a scale-free abstraction below the bridge-certificate threshold.

6 Main theorem

Theorem 6.1 (Minimum-viable MHL no-go theorem). *There exists a model of A_{MHL} and π in which*

$$F_q(\ell_q) \leq F_\Theta(\ell_\Theta)$$

is false. Therefore

$$A_{\text{MHL}} + \pi \not\models F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

7 Countermodel

We construct the model explicitly.

7.1 Pilot domains

Let

$$P_q = P_\Theta = \mathbb{R}_{\geq 0}.$$

Choose

$$p_q = p_\Theta = 1.$$

7.2 Monoids

Let

$$M_q = M_\Theta = (\mathbb{R}_{\geq 0}, +).$$

This is a nontrivial commutative monoid with identity 0.

7.3 Hulls

Let

$$H_q = H_\Theta = \{[0, t] : t \in \mathbb{R}_{\geq 0}\}.$$

Define

$$\text{Hull}_i(t) = [0, t].$$

7.4 Monoid maps

Define

$$\text{Mon}_i(p) = p.$$

Thus

$$\text{Mon}_q(p_q) = 1, \quad \text{Mon}_\Theta(p_\Theta) = 1.$$

7.5 Log-volume domains

Let

$$S_q = \mathbb{R}_q, \quad S_\Theta = \mathbb{R}_\Theta,$$

typed copies of $\mathbb{R}$.

7.6 Log-volume maps

Define

$$\text{LogVol}_q([0, t]) = -t \in S_q,$$

and

$$\text{LogVol}_\Theta([0, t]) = -2t \in S_\Theta.$$

Then

$$\ell_q = \text{LogVol}_q([0, 1]) = -1,$$

and

$$\ell_\Theta = \text{LogVol}_\Theta([0, 1]) = -2.$$

7.7 Common container

Let

$$C = \mathbb{R}$$

with

$$\preceq_C = \leq.$$

Define

$$\iota_q(x) = x$$

and

$$\iota_\Theta(y) = \frac{1}{4}y.$$

Then

$$\iota_q(\ell_q) = \iota_q(-1) = -1,$$

while

$$\iota_\Theta(\ell_\Theta) = \iota_\Theta(-2) = -\frac{1}{2}.$$

Therefore

$$\iota_q(\ell_q) \leq \iota_\Theta(\ell_\Theta)$$

is

$$-1 \leq -\frac{1}{2},$$

which is true. Hence the certificate π exists.

7.8 Numerical maps

Define

$$F_q(x) = x, \quad F_\Theta(y) = y.$$

Then

$$F_q(\ell_q) = F_q(-1) = -1,$$

and

$$F_\Theta(\ell_\Theta) = F_\Theta(-2) = -2.$$

The target conclusion becomes

$$-1 \leq -2,$$

which is false.

Thus the model satisfies $A_{\text{MHL}} + \pi$ but falsifies Num.

8 Verification of the countermodel

Lemma 8.1 (Typed domains exist). *The sets $S_q = \mathbb{R}_q$ and $S_\Theta = \mathbb{R}_\Theta$ are typed log-volume domains.*

Proof. They are stipulated typed copies of $\mathbb{R}$. □

Lemma 8.2 (The monoids are nontrivial commutative monoids).

$$M_q = M_\Theta = (\mathbb{R}_{\geq 0}, +)$$

are nontrivial commutative monoids.

Proof. Addition on $\mathbb{R}_{\geq 0}$ is associative and commutative, 0 is an identity, and $0 \neq 1$. □

Lemma 8.3 (The hull domains are nontrivial).

$$H_i = \{[0, t] : t \in \mathbb{R}_{\geq 0}\}$$

is nontrivial.

Proof. $[0, 0] \neq [0, 1]$. □

Lemma 8.4 (The pilot-to-monoid maps are well-defined).

$$\text{Mon}_i(p) = p$$

is well-defined.

Proof. Both P_i and M_i are $\mathbb{R}_{\geq 0}$. □

Lemma 8.5 (The hull maps are well-defined).

$$\text{Hull}_i(t) = [0, t]$$

is well-defined.

Proof. For every $t \in \mathbb{R}_{\geq 0}$, $[0, t] \in H_i$. □

Lemma 8.6 (The log-volume maps are well-defined).

$$\text{LogVol}_q([0, t]) = -t, \quad \text{LogVol}_\Theta([0, t]) = -2t$$

are well-defined.

Proof. For each $t \in \mathbb{R}_{\geq 0}$, both $-t$ and $-2t$ are real numbers in the corresponding typed copy. $\square$

Lemma 8.7 (The pilot-generated elements are -1 and -2).

$$\ell_q = -1, \quad \ell_\Theta = -2.$$

Proof.

$$\ell_q = \text{LogVol}_q(\text{Hull}_q(\text{Mon}_q(1))) = \text{LogVol}_q([0, 1]) = -1.$$

Likewise,

$$\ell_\Theta = \text{LogVol}_\Theta([0, 1]) = -2.$$

$\square$

Lemma 8.8 (The container comparison holds).

$$\iota_q(\ell_q) \leq \iota_\Theta(\ell_\Theta).$$

Proof.

$$\iota_q(\ell_q) = -1, \quad \iota_\Theta(\ell_\Theta) = -\frac{1}{2}.$$

Since

$$-1 \leq -\frac{1}{2},$$

the comparison holds. $\square$

Lemma 8.9 (The numerical conclusion fails).

$$F_q(\ell_q) \leq F_\Theta(\ell_\Theta)$$

is false.

Proof.

$$F_q(\ell_q) = -1, \quad F_\Theta(\ell_\Theta) = -2.$$

Thus the conclusion is

$$-1 \leq -2,$$

which is false. $\square$

Proof of the main theorem. By the lemmas above, the countermodel satisfies all axioms of A_{MHL} and the comparison certificate π . The same model falsifies Num. Hence

$$A_{\text{MHL}} + \pi \not\models \text{Num}.$$

$\square$

9 Parametric strengthening

The one-point countermodel is enough, but a parametric theorem is more informative.

Theorem 9.1 (Parametric active-rescaling no-go). *Let $a, b > 0$ with $a < b$. Set*

$$\ell_q = -a, \quad \ell_\Theta = -b.$$

Let

$$F_q(x) = x, \quad F_\Theta(y) = y.$$

Then the numerical conclusion

$$F_q(\ell_q) \leq F_\Theta(\ell_\Theta)$$

is

$$-a \leq -b,$$

which is false.

Choose any

$$0 < \lambda \leq \frac{a}{b}.$$

Define

$$\iota_q(x) = x, \quad \iota_\Theta(y) = \lambda y.$$

Then

$$\iota_q(\ell_q) = -a, \quad \iota_\Theta(\ell_\Theta) = -\lambda b.$$

Since $\lambda b \leq a$, we have

$$-a \leq -\lambda b.$$

Thus the container comparison holds while the numerical conclusion fails.

Proof. The proof is the calculation in the statement. □

Corollary 9.2 (Necessity of rescaling-breaking structure). *Any abstraction invariant under independent active positive rescaling of the Θ -side container representation, while holding F_Θ fixed, cannot entail the numerical conclusion.*

Proof. Apply the parametric active-rescaling theorem. If the abstraction is invariant under the active rescaling, then the rescaled model remains a model of the premises while falsifying the conclusion. □

10 Bridge-certificate insufficiency

Corollary 10.1 (No uniformly derivable bridge certificate in A_{MHL}). *No uniformly derivable bridge certificate B exists in A_{MHL} such that, in all A_{MHL} -models,*

$$\pi \Rightarrow R_B(\ell_q, \ell_\Theta, \pi)$$

and

$$R_B(\ell_q, \ell_\Theta, \pi) \Rightarrow F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

Proof. If such a uniformly derivable B existed, then by the conditional bridge theorem,

$$A_{\text{MHL}} + \pi \models F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

This contradicts the main theorem. □

Remark 10.2. *This corollary should be read carefully. It does not prohibit adding new bridge data to the language. It says only that A_{MHL} itself does not uniformly derive such a bridge.*

11 Escape hatches

The countermodel is defeated by adding any complete bridge mechanism. These additions are not flaws; they are precisely the missing data.

11.1 Chart bridge

Add a map

$$\kappa : C_{\text{pilot}} \rightarrow \mathbb{R}$$

such that κ is order-preserving and

$$F_q(x) = \kappa(\iota_q(x)), \quad F_\Theta(y) = \kappa(\iota_\Theta(y)).$$

Then the container comparison implies the numerical inequality.

In the countermodel, no such local order-preserving chart exists on the relevant two points, because it would require

$$\kappa(-1) = -1, \quad \kappa(-1/2) = -2,$$

contradicting order preservation.

11.2 Transport bridge

Add

$$g : S_q \rightarrow S_\Theta$$

such that

$$F_\Theta(gx) = F_q(x)$$

and

$$\pi \Rightarrow F_\Theta(g\ell_q) \leq F_\Theta(\ell_\Theta).$$

Then the numerical inequality follows.

11.3 Certified relation bridge

Add a relation

$$R_B(x, y, \rho)$$

such that

$$\rho \Rightarrow R_B(x, y, \rho)$$

and

$$R_B(x, y, \rho) \Rightarrow F_q(x) \leq F_\Theta(y).$$

This is the most general bridge form.

11.4 Interval-valued quotient bridge

Add

$$K : C_{\text{pilot}} \rightarrow \text{Intervals}(\mathbb{R})$$

with membership

$$F_q(x) \in K(\iota_q(x)), \quad F_\Theta(y) \in K(\iota_\Theta(y)),$$

and separation

$$\sup K(\iota_q(x)) \leq \inf K(\iota_\Theta(y)).$$

Then the numerical inequality follows.

11.5 Sandwich bridge

Add

$$\Phi_q : S_q \rightarrow \mathbb{R}, \quad \Phi_\Theta : S_\Theta \rightarrow \mathbb{R}$$

such that

$$F_q(x) \leq \Phi_q(x) \leq \Phi_\Theta(y) \leq F_\Theta(y).$$

11.6 Bounded-distortion-plus-margin bridge

Add ideal quantities

$$A_q, A_\Theta$$

and data

$$\delta > 0, \quad \varepsilon_q, \varepsilon_\Theta \geq 0$$

such that

$$\begin{aligned} A_q(x) + \delta &\leq A_\Theta(y), \\ F_q(x) &\leq A_q(x) + \varepsilon_q, \\ A_\Theta(y) - \varepsilon_\Theta &\leq F_\Theta(y), \end{aligned}$$

and

$$\varepsilon_q + \varepsilon_\Theta \leq \delta.$$

Then the numerical inequality follows.

11.7 Normalized monoid/hull valuation

Add a valuation

$$V_i : H_i \rightarrow \mathbb{R}$$

such that $F_i(\ell_i)$ factors through V_i , and π implies the valuation comparison.

11.8 Proof-path rank

Add a proof-certificate-dependent rank

$$\text{RankPath}(\rho)$$

whose extracted inequalities imply the numerical conclusion.

11.9 Calibration plus comparison-realizer

Add scale calibration, such as units, origins, or lattices, and pair it with a comparison-realizer. Calibration alone blocks some rescalings but does not by itself extract the final inequality.

12 Passive gauge versus active distortion

The countermodel uses *active* distortion. The numerical evaluation F_Θ remains fixed, while the container embedding uses

$$\iota_\Theta(y) = \frac{1}{4}y.$$

A passive gauge would transform numerical evaluation contravariantly so that numerical truth is preserved. Passive gauges are not the target of the countermodel. The no-go theorem targets abstractions that allow active relative rescaling without a bridge certificate.

13 Relevance to the IUT III bridge

The relevance is local and conditional.

The theorem does not model actual IUT. It says that a scale-free abstraction consisting only of monoid/hull/log-volume/common-container data is insufficient to obtain a numerical inequality. Therefore, any formalization of a Theorem 3.11-to-Corollary 3.12-like bridge must contain additional bridge data. This is aligned with the public formalization focus on Theorem 3.11 $\Rightarrow$ Corollary 3.12 and with the ordered-real-copy concern highlighted by Scholze–Stix [1, 2].

A careful relevance statement is:

This no-go theorem does not adjudicate IUT. It supplies a diagnostic lower bound: a formalization of the local common-container-to-numerical step must include a bridge certificate beyond scale-free monoid/hull/log-volume comparison data.

14 Limitations

- (i) A_{MHL} is not a faithful model of IUT.
- (ii) The countermodel is deliberately simple.
- (iii) The theorem applies only to scale-free abstractions.
- (iv) Stronger abstractions containing bridge certificates can evade the theorem.
- (v) The result is methodological, not arithmetic-geometric.
- (vi) The theorem does not address the full content of multiradial algorithms, Hodge theaters, Frobenioids, or theta-links.
- (vii) The theorem does not imply that actual IUT lacks the required bridge certificate.

15 Conclusion

We proved a minimum-viable no-go theorem:

$$A_{\text{MHL}} + \pi \not\leq F_q(\ell_q) \leq F_\Theta(\ell_\Theta).$$

The proof is by an active relative rescaling countermodel with nontrivial monoid, hull, and log-volume data. The common-container comparison holds, but the final numerical inequality fails.

The theorem shows that scale-free common-container comparison is insufficient. A successful bridge requires additional data: a chart, transport, certified relation, interval bound, sandwich estimate, bounded-distortion margin, normalized valuation, proof-path rank, or calibration plus comparison-realizer.

The modest contribution is a formal diagnostic for bridge formalization. If a proposed abstraction of a common-container-to-numerical step lacks all bridge-certificate mechanisms, then it cannot entail the desired numerical inequality.

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